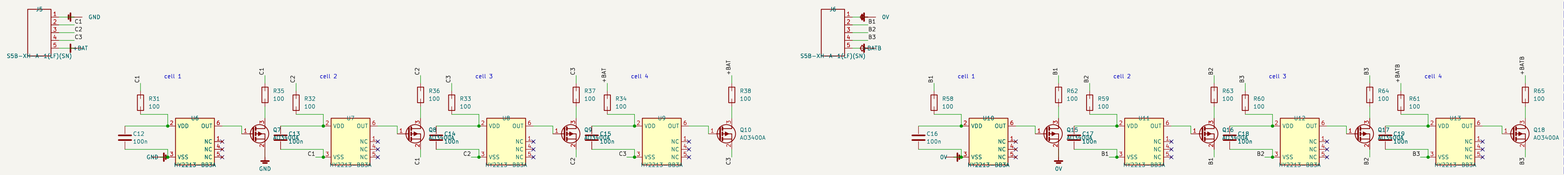
1. TWO 4S PACKS ON JST-XH BALANCE LEADS (pin 1 = black = pack negative) AND PER-CELL BALANCERS (HY2213-BB3A, 42 mA above 4.20 V)

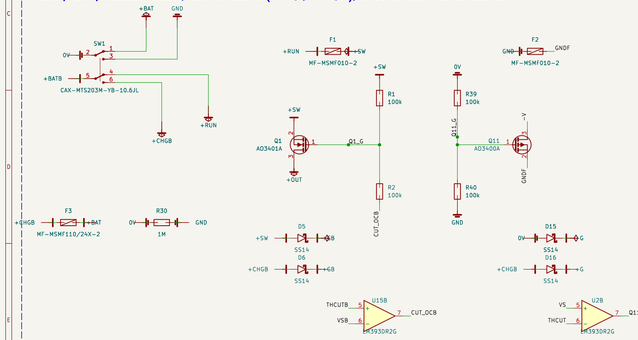
PACK A (−V side): GND .. +BAT



PACK B (+V side): 0V .. +BATB

Any 4S LiPo / Li-Ion hobby pack with a JST-XH 2.5 mm 5-pin balance plug (XT60 lead unused). Balancer per cell: HY2213 OUT drives an AO3400A that bleeds the cell through 100 R (42 mA, 0.18 W) above 4.200 V, off at 4.19 V; 2.5 uA standing. With the charger's 16.8 V this is the balance. Pack B's negative is the 0 V output node.

2. RUN / OFF / CHARGE TOGGLE, OUTPUT CHAINS (+V: Q1, −V: Q11), PARALLEL LINK FOR CHARGING



SW1 pole 1 (common 0V = pack B negative): RUN -> +BAT (packs in series, 0V = output 0 V); CHARGE -> GND (negatives joined). Pole 2 (common +BATB):

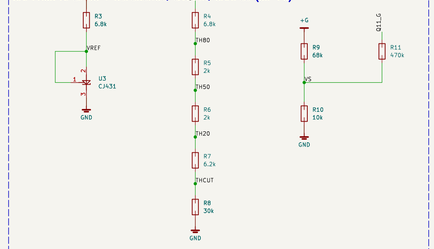
RUN -> +RUN -> F1 -> +SW -> Q1 -> +OUT (+V); CHARGE -> +CHGB -> F3 (1.1 A PTC) -> +BAT (positives joined: both packs on the one charger). OFF opens both.

R30 holds 0V at GND. Q1: U15B pulls the gate low (R1/R2) while VSB > THCUTB. Q11 (N, on pack A's negative): R39 from 0V and R40 give Vgs = +BAT/2 in RUN:

U2B pulls the gate low below 12.0 V (R11 470k from the gate; back on at 12.9 V): in OFF and CHARGE 0V is at GND, so Q11 is off and −V floats. Gauge rails:

+GB = +5W or +CHGB (D5/D6), +G = 0V or +CHGB (D15/D16): both gauges run in RUN and CHARGE and are dead in OFF.

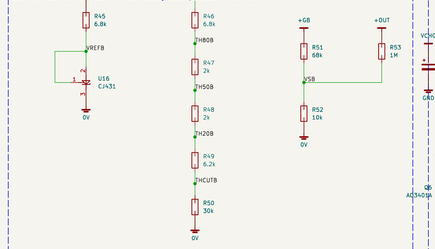
3A. PACK A: 2.495 V REFERENCE, LADDER, SAMPLE (on +G)



Ladder (107 uA): TH80 2.137 V, TH50 2.029 V, TH20 1.920 V, THCUT 1.595 V. VS = rail x 0.1371 with the output live, x 0.1271 cut off: rail = pack − 0.25 V (SS14).

Thresholds 16.0 / 15.2 / 14.4 V (pack) for the 80 / 50 / 20 % LEDs, cutoff 12.0 V, re-enable 12.8 V (pack B, R53 from +OUT) / 12.9 V (pack A, R11 from Q11's gate).

3B. PACK B: THE SAME ON +GB, REFERENCED TO 0V



5. OUTPUT AND MOUNTING HOLES



OUT: +V / 0V / −V / nc (JST-PH 4, to the module)

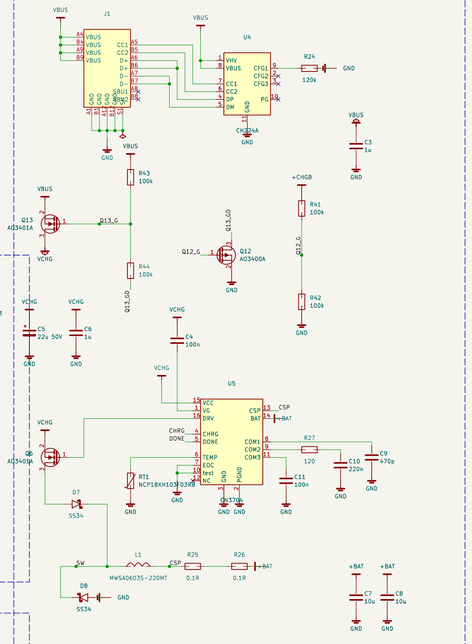


Packs strapped under the board (two 20 mm straps through the slots), leads to J5 / J6 at the front edge.

Toggle RUN powers the module; OFF; CHARGE joins the packs in parallel on the charger and opens both outputs. −V floats whenever pack A is off (OFF, CHARGE, under-voltage), +V whenever pack B is: the module's fault comparator reports a missing rail.

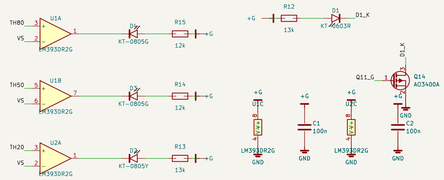
Standing draw in OFF about 40 uA (balancers, charger output stage).

6. USB-C PD 20 V INPUT (CH224A), VBUS SWITCH (CHARGE ONLY), 4S BUCK CHARGER (CN3704, 1 A)



J1 / U4: the CH224A asks a PD brick for 20 V (R24 120k on CFG1). Q13 passes VBUS to the charger only while Q12 sees +CHGB (toggle in CHARGE); a brick plugged in while RUN or OFF does nothing. U5: buck at 300 kHz, Q8 switch, D7 blocks the packs from the input, D8 freewheels. L1 22 uH. R25 + R26 = 0.2 R -> 1 A constant current shared by the two packs in parallel (F3). 16.8 V +-1 % constant voltage, termination at 9 % (92 mA), trickle at 150 mA below 11.2 V, re-charge below 16.0 V, over-voltage cut at 18.1 V. RT1 10k NTC (board temperature; a 10k resistor disables it). D9 CHRG (red), D10 DONE (green). All of it on GND = pack A negative = the USB-C shell.

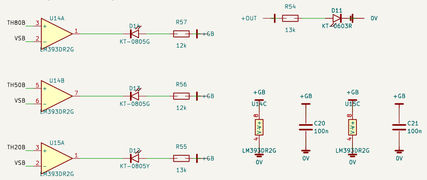
4A. PACK A GAUGE: THREE COMPARATORS, FOUR LEDs



D4 >= 16.0 V (80 %), D3 >= 15.2 V (50 %), D2 >= 14.4 V (20 %) from +G: D1 (ON) lit by Q14 while Q11 conducts (−V live).

C1 / C2 at pin 8 of U1 / U2. Bars read the open-circuit voltage (the module draws ~10 mA).

4B. PACK B GAUGE (on +GB / 0V)



D14 / D13 / D12 as pack A: D11 (ON) from +OUT = +V live. C20 / C21 at pin 8 of U14 / U15.

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Title: Battery board for the LaserHAT e-stim module: two 4S hobby packs, o

Size: A1	Date: 2026-10-27	Rev: C
KiCad E.D.A. 9.0.8		Id: 1/1